

Week6 Assignment 7

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Problem 1 Description

Consider the process transfer function for the Van de Vusse reactor

$$g_p(s) = \frac{0.5848(-0.3549s + 1)}{0.1858s^2 + 0.8627s + 1}$$

Find the Ziegler-Nichols controller parameters for P, PI, and PID controllers for this process, based on the closed-loop oscillation method. Compare the responses of all three controllers to a step setpoint change.

Solution

The closed-loop oscillation method is a PID digital control parameter tuning method based on the critical proportional method in analog controllers.

When tuning parameters using the closed-loop oscillation method, the first step is to determine the critical proportional gain k_{cu} and the critical oscillation period P_u .

The specific method is to input the sampling period T_{min} (default is 0.1s) into the computer control system and use pure proportional control. The proportional gain K_c is gradually decreased (i.e., the proportional gain K_c is increased) until the system produces sustained oscillations. The proportional gain K_c obtained at this point is the critical proportional gain k_{cu} , and the corresponding oscillation period is called the critical oscillation period P_u .

We use a trial-and-error approach and set the integral and derivative terms to zero ($K_i = 0(\tau_I = \infty)$, $K_d = 0(\tau_D = 0)$), then adjust K_c until sustained oscillations appear:

```
Ts3 = 1;
Gd3 = c2d(G, Ts3, 'zoh');

Kp_try = 5.18105;
Ki_try = 0;
Kd_try = 0;

PID_try = pid(Kp_try, Ki_try, Kd_try, 'Ts', Ts3);
sys_cl_try = feedback(PID_try * Gd3, 1);

N3 = 200;
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t3 = 0:Ts3:(N3-1)*Ts3;
[y_try, t_try] = step(sys_cl_try, t3);

figure;
stairs(t_try, y_try, 'b', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('System Response');
title('Trial-and-Error Method (Ts = 1s)');

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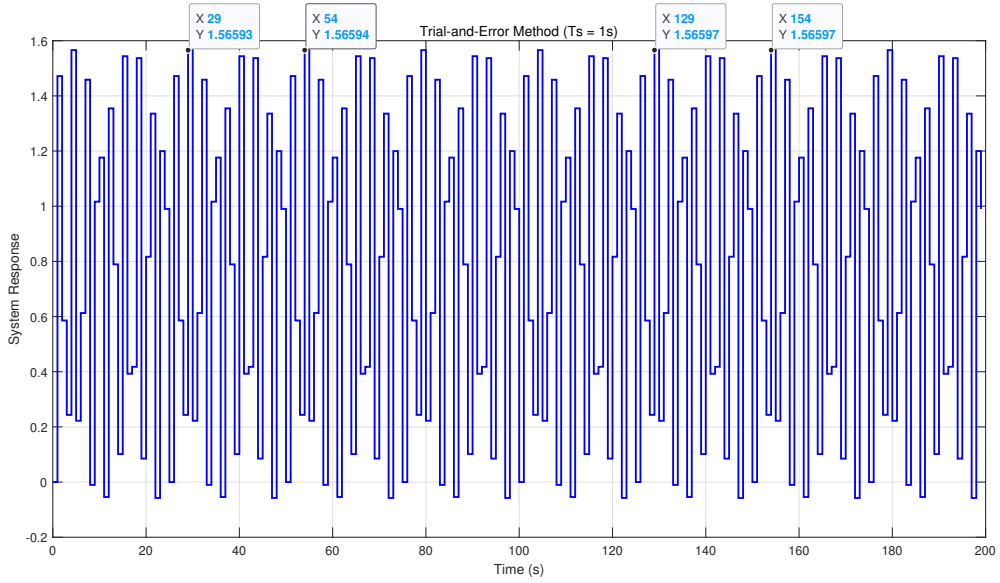


Fig. 1: k_{cu} & P_u

From Figure 1, it can be observed that the system produces sustained oscillations, and the proportional gain $K_p = 13.525$ is the value of k_{cu} , while the critical oscillation period is $P_u = 3.36$ s.

Subsequently, we use the Ziegler-Nichols conditions:

Controller type	K_c	τ_I	τ_D
P-only	$0.5k_{cu}$	—	—
PI	$0.45k_{cu}$	$P_u/1.2$	—
PID	$0.6k_{cu}$	$P_u/2$	$P_u/8$

Table. 1: Controller Parameters

We calculate the parameters and perform simulations as follows:

```

Ts1 = 2.9; Gd1 = c2d(G, Ts1, 'zoh'); k_cu = 5.18105; P_u = 29;
% P    Kp_ZN = 0.5 * k_cu;    Ki_ZN = 0;                    Kd_ZN = 0;
% PI    Kp_ZN = 0.45 * k_cu; Ki_ZN = 0.54 * k_cu / P_u; Kd_ZN = 0;
% PID   Kp_ZN = 0.6 * k_cu;  Ki_ZN = 1.2 * k_cu / P_u;    Kd_ZN = 0.075 * k_cu * P_u;

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PID_ZN = pid(Kp_ZN, Ki_ZN, Kd_ZN, 'Ts',Ts1);
sys_cl_ZN = feedback(PID_ZN * Gd1, 1);
N1 = 20;
t1 = 0:Ts1:(N1-1)*Ts1;
[y_ZN, t_ZN] = step(sys_cl_ZN, t1);
figure; stairs(t_ZN, y_ZN, 'r', 'LineWidth', 1.5);
grid on; xlabel('Time (s)'); ylabel('System Response');
title('Closed-loop oscillation method (Ts = 2.9s) ');

```

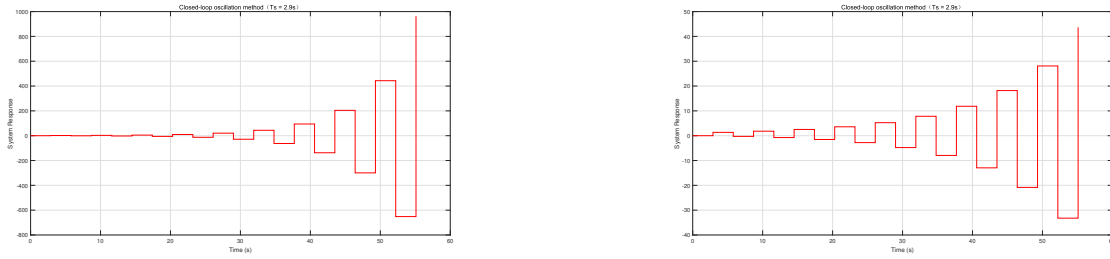


Fig. 2: Closed-loop oscillation method - P(a) & PI(b)

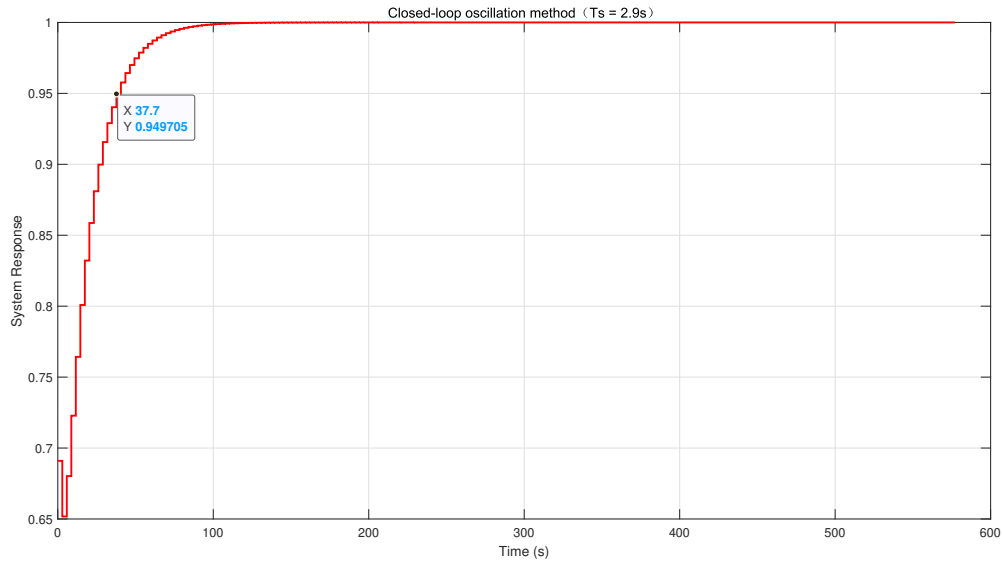


Fig. 3: Closed-loop oscillation method - PID

The system responses for the three controllers are shown in Figures 2 and 3. By comparison, it can be observed that the system is unstable with the P and PI controllers, while the PID controller results in better dynamic performance without overshoot. The rise/settle time is approximately 37.7s (95%). Since this simulation is for an ideal PID, a slight derivative kick can be observed at the beginning of the step input.

Problem 2 Description

Consider the following first-order process:

$$g_p(s) = \frac{2}{3s + 1}$$

If the desired closed-loop response to a setpoint change is second order with the following form,

$$g_{CL}(s) = \frac{\alpha s + 1}{(\lambda s + 1)^2}$$

Find the feedback controller required, where α and λ are adjustable tuning parameters (they are both positive). What type of controller is this? If the controller is PID form (perhaps with a lag), find each of the tuning parameters ($k_c, \tau_I, \tau_D, \tau_F$). Show that $\lambda > 0.5\alpha$ is required for the controller to be stable.

Solution

We are given the first-order process transfer function:

$$g_p(s) = \frac{2}{3s + 1}$$

and we want to design a feedback controller $G(s)$ such that the closed-loop transfer function $g_{CL}(s)$ has the following form:

$$g_{CL}(s) = \frac{\alpha s + 1}{(\lambda s + 1)^2}$$

where α and λ are positive adjustable tuning parameters.

Step 1: General Closed-Loop Transfer Function

The closed-loop transfer function with feedback is:

$$g_{CL}(s) = \frac{g_p(s)G_C(s)}{1 + g_p(s)G_C(s)}$$

Substituting $g_p(s) = \frac{2}{3s+1}$, we get:

$$g_{CL}(s) = \frac{\frac{2}{3s+1}G_C(s)}{1 + \frac{2}{3s+1}G_C(s)}$$

We are given that the desired closed-loop transfer function is:

$$g_{CL}(s) = \frac{\alpha s + 1}{(\lambda s + 1)^2}$$

Equating the two expressions for $g_{CL}(s)$, we obtain:

$$\frac{\frac{2}{3s+1}G_C(s)}{1 + \frac{2}{3s+1}G_C(s)} = \frac{\alpha s + 1}{(\lambda s + 1)^2}$$

Step 2: Solving for the Controller $G_C(s)$

$$\frac{2(\lambda s + 1)^2}{3s + 1}G_C(s) = \alpha s + 1 + \frac{2(\alpha s + 1)}{3s + 1}G_C(s)$$

$$G_C(s) = \frac{3\alpha s^2 + (3 + \alpha)s + 1}{2\lambda^2 s^2 + (4\lambda - 2\alpha)s}$$

Step 3: Type of Controller

The controller $G(s)$ is of a form that is typically classified as a PID controller with a derivative filter. To express it in PID form, we need to decompose it into proportional, integral, derivative, and possible filtering terms.

Step 4: Tuning Parameters

The PID controller in terms of k_c , τ_I , τ_D , and τ_F can be written as:

$$G_C(s) = k_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right) \frac{1}{1 + \tau_F s} = \frac{k_c \tau_I \tau_D s^2 + k_c \tau_I s + k_c}{\tau_I \tau_F s^2 + \tau_I s}$$

By comparing this form with the derived expression for $G(s)$, we can identify the tuning parameters.

$$G_C(s) = \frac{3\alpha s^2 + (3 + \alpha)s + 1}{2\lambda^2 s^2 + (4\lambda - 2\alpha)s} = \frac{k_c \tau_I \tau_D s^2 + k_c \tau_I s + k_c}{\tau_I \tau_F s^2 + \tau_I s}$$

$$k_c = 1 \quad \tau_I = 3 + \alpha = 4\lambda - 2\alpha \quad \tau_D = \frac{3\alpha}{\tau_I} \quad \tau_F = \frac{2\lambda^2}{\tau_I}$$

Step 5: Stability Condition

To ensure the stability of the system, we need to show that $\lambda > 0.5\alpha$ is required. This can be done by analyzing the root locus or performing a stability analysis using the Nyquist or Bode criteria.

From the transfer function of the controller $G_C(s)$, it is known that its characteristic equation is $\tau_I \tau_F s^2 + \tau_I s = 0$. Therefore, the characteristic roots are $\frac{-1}{\tau_F} = \frac{-\tau_I}{2\lambda^2}$. To ensure the stability of the controller, i.e., the characteristic roots are in the left half-plane, it is necessary to satisfy $\tau_I > 0$, which implies $3 + \alpha = 4\lambda - 2\alpha > 0$, thus $\alpha > -3$ and $\lambda > 0.5\alpha$. This completes the proof.